

Summary of the Non-Diagonal-Mass Construction Path

SphericalSBPOperators Development Team¹

¹*Computational Science and Engineering*

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This note summarizes the current `src/non_diagonal_mass/` construction path in `SphericalSBPOperators.jl`. The emphasis is on the ansatz for the scalar mass S and vector mass V , and on the conditions used in `sbp4_solve_accuracy_constraints`, `sbp6_solve_accuracy_constraints`, and `sbp8_solve_accuracy_constraints` to determine them.

I. COMMON CONSTRUCTION PATH

All three non-diagonal-mass paths follow the same top-level sequence:

1. Build a full-grid Cartesian SBP operator on $[-R, R]$.
2. Fold it to the half-grid $r \in [0, R]$ using even/odd parity to obtain

$$G_{\text{even}} = R_{\text{op}} G_{\text{full}} E_{\text{even}}, \quad G_{\text{odd}} = R_{\text{op}} G_{\text{full}} E_{\text{odd}}. \quad (1)$$

3. Form the folded Cartesian half-mass

$$H_{\text{cart,half}} = \frac{1}{2} E_{\text{even}}^T H_{\text{full}} E_{\text{even}}. \quad (2)$$

4. Use the metric weight r^p to define the diagonal seed

$$S_{\text{seed}} = H_{\text{cart,half}} \text{diag}(r^p). \quad (3)$$

5. Solve for a diagonal scalar mass S and a structured symmetric vector mass V , then define the divergence through the SBP identity

$$D = S^{-1}(B - G_{\text{even}}^T V), \quad B = \text{diag}(0, \dots, 0, r_N^p). \quad (4)$$

The public constructors `sbp4_operators`, `sbp6_operators`, and `sbp8_operators` all return

$$(D, G, S, V, B), \quad G = G_{\text{even}}. \quad (5)$$

II. WHAT IS ASSUMED FOR S AND V ?

A. Assumptions shared by all orders

- S is always diagonal.
- V is always symmetric.
- The non-diagonal part of V is localized near the origin.
- Away from the origin, the code ties the tail of V back to S through $V_{ii} = S_{ii}$ once the boundary block ends.

So the entire method can be read as:

Keep S simple, keep the far field diagonal, and put the non-diagonal freedom only in a small origin block of V .

B. SBP4 ansatz

The symbolic template in `sbp4_vector_mass` assumes

$$S = \text{diag}(s_1, \dots, s_N). \quad (6)$$

For V :

$$V_{11} = 1, \quad (7)$$

$$V_{22}, V_{33}, V_{44}, V_{55} \text{ are free}, \quad (8)$$

$$V_{23} = V_{32}, \quad V_{34} = V_{43}, \quad (9)$$

$$V_{45} = V_{54} \text{ is the last allowed off-diagonal pair}, \quad (10)$$

$$V_{ii} = s_i \quad \text{for } i \geq 6. \quad (11)$$

Thus only the leading 5×5 block can differ from the diagonal seed.

C. SBP6 ansatz

The symbolic template in `sbp6_vector_mass` keeps

$$S = \text{diag}(s_1, \dots, s_N), \quad (12)$$

but expands the admissible boundary block for V :

$$V_{11} = 1, \quad (13)$$

$$V_{ii} \text{ are free for } i = 2, \dots, \min(10, N), \quad (14)$$

$$V_{ii} = s_i \quad \text{for } i \geq 11. \quad (15)$$

The allowed off-diagonal pairs are

$$\begin{aligned} &(2, 3), (2, 4), \\ &(3, 4), (3, 5), \\ &(4, 5), (4, 6), \\ &(5, 6), (5, 7), \\ &(6, 7), \end{aligned} \quad (16)$$

whenever those indices exist. So the SBP6 V block is diagonal in the tail and pentadiagonal-like near the origin.

D. SBP8 ansatz

For SBP8 the structure is more flexible. The template in `sbp8_vector_mass` assumes

$$S = \text{diag}(s_1, \dots, s_N), \quad (17)$$

and lets V have a banded symmetric boundary block:

$$V_{ii} \text{ free for } i \leq n_b, \quad (18)$$

$$V_{ii} = s_i \text{ for } i > n_b, \quad (19)$$

with off-diagonals allowed only inside the leading $n_b \times n_b$ block and only up to a chosen odd bandwidth. The default exact path uses a heptadiagonal pattern (`boundary_bandwidth = 7`).

Unlike SBP4 and SBP6, SBP8 does not hard-wire $V_{11} = 1$. The origin diagonal of V is solved together with the rest of the boundary block.

III. WHAT CONDITIONS ARE USED TO SOLVE FOR S AND V ?

A. Shared algebraic condition

All three orders use the same discrete divergence identity:

$$SD + G_{\text{even}}^T V = B. \quad (20)$$

Applied to an odd monomial $u(r) = r^d$, this gives the row-wise condition

$$-(G_{\text{even}}^T V u)_i - s_i(p+d)r_i^{d-1} + B_{ii}u_i = 0. \quad (21)$$

This is linear in the unknown entries of S and V , and it is the core equation used to populate the solve matrix in every order.

All three orders also impose the scalar quadrature condition

$$\sum_{i=1}^N s_i = \int_0^R r^p dr = \frac{R^{p+1}}{p+1}. \quad (22)$$

B. SBP4 conditions

The exact SBP4 solve only frees the boundary block. The unknown vector is

$$x = [s_1, \dots, s_5, v_2, v_3, v_4, v_5, v_{23}, v_{34}, v_{45}]^T, \quad (23)$$

while for $i \geq 6$ the scalar tail is fixed to

$$s_i = (S_{\text{seed}})_{ii}, \quad V_{ii} = s_i. \quad (24)$$

The solver then enforces:

$$Dr = (p+1) \text{ on rows } 1 : (N-4), \quad (25)$$

$$Dr^3 = (p+3)r^2 \text{ on rows } 1 : \min(5, N), \quad (26)$$

$$\sum_i s_i = \frac{R^{p+1}}{p+1}. \quad (27)$$

So SBP4 assumes that only the first five scalar masses and the first few vector-mass entries need to move; the far field is inherited directly from the folded diagonal seed.

C. SBP6 conditions

The SBP6 exact solve frees the entire diagonal of S :

$$x = [s_1, \dots, s_N, v_2, \dots, v_{\min(10, N)}, \{v_{ij}\}_{(i,j) \in \mathcal{P}_6}]^T, \quad (28)$$

where $\mathcal{P}_6$ is the allowed off-diagonal pair list above. For $i \geq 11$, the vector-mass diagonal is tied to the solved scalar mass:

$$V_{ii} = s_i. \quad (29)$$

The row sets are chosen from the inferred right-boundary closure width `closure_right`, and the solver enforces:

$$Dr = (p+1) \text{ on all rows}, \quad (30)$$

$$Dr^3 = (p+3)r^2 \text{ on rows } 1 : (N - \text{closure_right}), \quad (31)$$

$$Dr^5 = (p+5)r^4 \text{ on rows } 1 : \min(\text{first_rows_r5}, N), \quad (32)$$

$$\sum_i s_i = \frac{R^{p+1}}{p+1}. \quad (33)$$

Relative to SBP4, SBP6 relaxes much more of the diagonal structure. The code is effectively assuming that sixth order needs a wider and more global redistribution of scalar weights.

D. SBP8 exact conditions

The preferred SBP8 path is the exact rational solve in `_sbp8_solve_accuracy_constraints_exact`. It frees:

- all scalar masses $s_1, \dots, s_N$,
- the selected leading vector diagonal entries V_{ii} for $i \in \{1, \dots, v_{\text{diag, end}}\}$,
- the allowed boundary-block off-diagonals inside the banded pattern.

Outside that free diagonal set, the code ties V_{ii} back to s_i .

The enforced polynomial conditions are

$$Dr = (p+1) \text{ on all rows}, \quad (34)$$

$$Dr^3 = (p+3)r^2 \text{ on rows excluding the right closure}, \quad (35)$$

$$Dr^5 = (p+5)r^4 \text{ on rows excluding the right closure}, \quad (36)$$

$$Dr^7 = (p+7)r^6 \text{ on rows } 1 : \min(\text{first_rows_r7}, N), \quad (37)$$

$$\sum_i s_i = \frac{R^{p+1}}{p+1}. \quad (38)$$

The exact SBP8 solve also adds an origin anchor

$$s_1 = s_{1,\text{target}} \quad (39)$$

to avoid degenerate exact solutions with $S_{11} = 0$. By default this target is chosen from the local grid spacing, approximately $0.55 h^3$ when $p = 2$.

IV. EXTRA ADMISSIBILITY CONDITIONS IN THE SBP8 PATH

SBP8 is the only order that has a second solve path if the exact solve fails. The default public function tries the exact solve first and falls back to a JuMP/Ipopt nonlinear program.

In the fallback path, the equality constraints are the same divergence and quadrature conditions as above, but the code additionally enforces:

1. positivity bounds

$$s_i \geq s_{\min}, \quad V_{ii} \geq s_{\min} \quad (40)$$

on the free boundary variables;

2. positive definiteness of the boundary block of V through a Cholesky factorization

$$V_{\text{boundary}} = LL^T; \quad (41)$$

3. a small objective that keeps S near the folded seed and keeps V close to S .

After either solve path, SBP8 also checks:

- positivity of the diagonal of S ,

- positive definiteness of the boundary block of V ,
- optionally positive definiteness of the full V ,
- floating-point level satisfaction of the enforced constraints,
- optionally that the spectrum of $L = DG$ is real and non-positive to within tolerance.

V. TAKEAWAY

The non-diagonal-mass path can be summarized in one sentence:

Build a folded half-grid gradient, keep the scalar mass diagonal, allow only a compact origin-local non-diagonal correction in the vector mass, and determine the free coefficients by enforcing odd-monomial divergence exactness plus one scalar quadrature constraint.

Order by order, the main difference is how much freedom is granted:

- SBP4: only the first five scalar masses and a tiny V boundary block move.
- SBP6: all scalar masses move, with a wider pentadiagonal-like V boundary block.
- SBP8: all scalar masses move, the V boundary block becomes general banded, and the code adds an origin anchor plus stronger admissibility checks.